

Do **not** write solutions on this page.

12. [Maximum mark: 19]

- (a) Solve $z^2 = -1 - \sqrt{3}i$, giving your answers in the form $z = r(\cos \theta + i \sin \theta)$. [4]

Let z_1 and z_2 be the square roots of $-1 - \sqrt{3}i$, where $\operatorname{Re}(z_1) > 0$.

Let z_3 and z_4 be the square roots of $-1 + \sqrt{3}i$, where $\operatorname{Re}(z_3) > 0$.

- (b) Expressing your answers in the form $z = a + bi$, where $a, b \in \mathbb{R}$,

(i) find z_1 and z_2 ;

(ii) deduce z_3 and z_4 . [4]

The four roots z_1, z_2, z_3 and z_4 are represented by the points A, B, C and D respectively on an Argand diagram.

- (c) (i) Plot the points A, B, C and D on an Argand diagram.

(ii) Find the area of the polygon formed by these four points. [4]

The four roots z_1, z_2, z_3 and z_4 satisfy the equation $z^4 + 2z^2 + 4 = 0$.

The four roots $\frac{1}{z_1}, \frac{1}{z_2}, \frac{1}{z_3}$ and $\frac{1}{z_4}$ satisfy the equation $pw^4 + qw^2 + r = 0$ where $p, q, r \in \mathbb{Z}$.

- (d) Find the value of p, q and r . [3]

The four roots $\frac{1}{z_1}, \frac{1}{z_2}, \frac{1}{z_3}$ and $\frac{1}{z_4}$ are represented by the points E, F, G and H respectively on an Argand diagram.

- (e) (i) Find $\frac{1}{z_1}$ in the form $z = a + bi$, where $a, b \in \mathbb{R}$.

(ii) Hence, deduce the area of the polygon formed by these four points. [4]

